

Experiment: Integration of Information Retrieval with Knowledge Graphs

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Quiz Evaluation:	1 / 10 (10%)	Status:	Completed

1. Aim & Learning Objectives

- Aim:** Combine document retrieval results with graph-based entity and relationship exploration.
- Index heterogeneous documents (PDF, DOCX, TXT, HTML, PPTX) into a Vector Space Model using TF-IDF.
 - Calculate Cosine Similarity to score, rank, and retrieve relevant documents based on user query.
 - Perform offline rule-based entity extraction and frequency analysis across retrieved texts.
 - Synthesize semantic relationships into an interactive directed Knowledge Graph with NetworkX.
 - Explore multi-relational concept paths to evaluate conceptual overlap and domain connections.

2. Latest Information Retrieval & Graph Generation Execution

Active Query: "Types of AI and their use in Science"

Parameters: Top-K = 5 | Cosine Threshold = 0.05

Rank	Document Name	Type	Similarity
1	doc10_information_retrieval_tfidf.txt	TXT	10.73%
2	doc01_ai_fundamentals.txt	TXT	9.15%

Extracted Entities: Artificial Intelligence (2), Documents (2), AI (1), Machine Learning (1), Data (1), Algorithm (1), Information Retrieval (1), TF-IDF (1)

Knowledge Graph Edges: (Artificial Intelligence -[uses]-> Machine Learning); (TF-IDF -[retrieves]-> Documents)

3. Experimental Trials Record Table

Trial #	Timestamp	Query	Top K	Threshold	Retrieved Coun	Average Simila	Entity Count	Relationship C
1	23:28:57	Types of AI an	5	0.05	2	9.94%	8	2

4. Experimental Observations

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5. Conclusion

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Student / Faculty Evaluator Signature